

Biologically meaningful distribution models highlight the benefits of the Paris Agreement for demersal fishing targets in the North Atlantic Ocean

Manuel Ramos Martins  | Jorge Assis  | David Abecasis 

Centre of Marine Sciences (CCMAR),
University of the Algarve, Faro, Portugal

Correspondence

Manuel Ramos Martins, Centre of Marine
Sciences (CCMAR), University of the
Algarve, Campus de Gambelas, 8005-139
Faro, Portugal.
Email: mrmartins@ualg.pt

Funding information

MAR2020; European Maritime and
Fisheries Fund; Fundação para a Ciência e a
Tecnologia

Editor: Aaron MacNeil

Abstract

Aim: With climate change challenging marine biodiversity and resource management, it is crucial to anticipate future latitudinal and depth shifts under contrasting global change scenarios to support policy-relevant biodiversity impact assessments [e.g., Intergovernmental Panel on Climate Change (IPCC)]. We aim to demonstrate the benefits of complying with the Paris Agreement (United Nations Framework Convention on Climate Change) and limiting environmental changes, by assessing future distributional shifts of 10 commercially important demersal fish species.

Location: Northern Atlantic Ocean.

Time period: Analyses of distributional shifts compared near present-day conditions (2000–2017) with two Representative Concentration Pathway (RCP) scenarios of future climate changes (2090–2100): one following the Paris Agreement climate forcing (RCP2.6) and another without stringent mitigation measures (RCP8.5).

Major taxa studied: Demersal fish.

Methods: We use machine learning distribution models coupled with biologically meaningful predictors to project future latitudinal and depth shifts. Structuring projections with information beyond temperature-based predictors allowed us to encompass the physiological limitations of species better.

Results: Our models highlighted the additional roles of temperature, primary productivity and dissolved oxygen in shaping fish distributions (average relative contribution to the models of 32.12 ± 10.24 , 15.6 ± 7.5 and $12.1 \pm 6.1\%$, respectively). We anticipated a generalized trend of poleward shifts in both future scenarios, with aggravated changes in suitable area with RCP8.5 (average area loss with RCP2.6 = $13.3 \pm 4.1\%$; RCP8.5 = $40.9 \pm 13.3\%$). Shifts to deeper waters were also predicted to be of greater magnitude with RCP8.5 (average depth gain = 25.4 ± 21.5 m) than with RCP2.6 (average depth gain = 10.4 ± 7.9 m). Habitat losses were projected mostly in the Mediterranean, Celtic and Irish Seas, the southern areas of the North Sea and along the NE coast of North America.

Main conclusions: Inclusion of biologically meaningful predictors beyond temperature in species distribution modelling can improve predictive performances. Limiting future climate changes by complying with the Paris Agreement can translate into

reduced distributional shifts, supporting biodiversity conservation and resource management.

KEYWORDS

biologically meaningful predictors, climate change, demersal fish, North Atlantic fisheries, Paris agreement, species distribution modelling

1 | INTRODUCTION

Empirical evidence shows climate-driven changes producing poleward and depth shifts across marine taxa (Pinsky et al., 2020). Future changes in physical (e.g., temperature), chemical (e.g., oxygen content) and biological (e.g., primary productivity) ocean components, linked to greenhouse gas emissions, are expected to shift the distribution of species and ecosystems further (Champion et al., 2018; Cheung et al., 2010), with severe implications for marine biodiversity and fisheries (Poloczanska, 2013; Sumaila et al., 2011; Weatherdon et al., 2016). On the one hand, local population extinctions at low-latitude distributional margins might produce negative cascading effects and reduce global genetic pools (Cheung et al., 2013). On the other hand, expansions to higher latitudes might change biological interactions (e.g., competition) and restructure the composition of biological communities (Pech et al., 2017).

To address potential changes in a timely manner, the Paris Agreement was developed to provide a roadmap limiting global warming to $< 2^{\circ}\text{C}$ above pre-industrial levels while striving to contain the increase to 1.5°C (United Nations Framework Convention on Climate Change, 2015). This initiative considered climate change impact assessments on biodiversity [e.g., Intergovernmental Panel on Climate Change (IPCC)] based on policy-relevant research (Geden, 2016). Such assessments are mostly performed with species distribution models (SDMs) contrasting near present-day distributions with projections derived from atmosphere–ocean general circulation models (AOGCMs), which estimate climate conditions with different concentrations of greenhouse gases and mitigation strategies (Handorf & Dethloff, 2012; Melo-Merino et al., 2020). Species distribution modelling has the advantage of allowing ecological applications across taxa and systems; however, models that do not incorporate the broad complexity of environmental conditions driving geographical distributions could yield misleading impact assessments (McHenry et al., 2019; Spence & Tingley, 2020). For instance, marine SDMs generally miss important drivers beyond thermal conditions, even though the distribution of organisms is influenced by additional drivers, such as dissolved oxygen, salinity, primary productivity and nutrients (Duncan et al., 2020; Pörtner & Farrell, 2008; Pörtner & Knust, 2007; Sunday et al., 2012).

Here, we produce a policy-relevant biodiversity impact assessment anticipating future distributional shifts of 10

demersal fish species with high commercial value in the intensively exploited northern Atlantic Ocean (EU, 2018; FAO Fisheries Department, 2011). In this broad area, climate conditions are changing significantly (Cheung et al., 2010; Free et al., 2019) and producing distributional shifts for demersal fish, both latitudinally and in depth (Mills et al., 2013; Perry et al., 2005). For instance, exposure to oxygen minimum zones (OMZ) is escalating in the northern Atlantic, with most of the anoxic regions of the world's (Breitburg et al., 2018) expanding to coastal areas where fishing targets reside (Diaz & Rosenberg, 2008). In the same way, the expected changes in primary productivity and nutrient availability might also produce a challenging environment for fish to thrive (Cheung et al., 2010). Accordingly, fisheries management might be less effective in the face of environmental changes (Baudron et al., 2020), with conflicts over catch allocation as fishing targets shift across management boundaries, which can lead to the discard of high fished volumes owing to poor allocation of fish stocks based on outdated species distribution information (Bell et al., 2015; Miller & Munro, 2004).

Previous studies striving to anticipate future distributions of fish species in the northern Atlantic Ocean have mostly been focused on single species and largely neglected additional predictors, such as oxygen availability or productivity changes (Kleisner et al., 2017; Rutterford et al., 2015). We use machine learning SDM, fitting a comprehensive dataset of species occurrence records with biologically meaningful predictors (i.e., predictors that are known to have a direct impact on species physiology, and therefore, potentially structure distributional limits; Rödder & Lötters, 2010) tailored for demersal conditions. Near present-day distributional patterns are compared with two scenarios of future climate change: the Representative Concentration Pathway 2.6 (RCP2.6; van Vuuren et al., 2011), following the Paris Agreement expectations of global warming levels $< 2^{\circ}\text{C}$ (Wang et al., 2018), and the Representative Concentration Pathway 8.5 (RCP8.5; Riahi et al., 2011), with minimal mitigation measures and greater climatic changes. By comparing contrasting future scenarios, we aim to provide key information for IPCC impact assessments, for the fulfilment of the Sustainable Development Goals of the 2030 Agenda (United Nations General Assembly, 2015) and for better-informed conservation and management. Our hypotheses are as follows: (1) additional drivers beyond thermal conditions are key to modelling the distribution of demersal fish, and (2) latitudinal and depth shifts are positively correlated with anthropogenic greenhouse

gas emissions, meaning that there is a substantial benefit to complying with the Paris Agreement (as inferred for terrestrial climate; Warren et al., 2018), limiting global changes and, consequently, biodiversity losses.

2 | METHODS

2.1 | Focal species and study area

We modelled near present-day and future distributions of 10 demersal fish species (Table 1) throughout the northern Atlantic Ocean and adjacent seas (i.e., the Mediterranean Sea, the Black Sea and the Barents Sea; study area extending from 120°E to 120°W and from 30 to 90°N). In this region, fishing gears targeting demersal species (demersal purse seiners and trawlers) are more prominently present in the north-east (Figure 1b). For this reason, we focused our study on species that are concentrated geographically on the eastern side of the ocean, while still having high commercial value in the whole Atlantic (EU, 2018; FAO Fisheries Department, 2011). Focal species have different depth ranges, which were considered while modelling distributions (Table 1), and are exclusively demersal, with the exceptions of *Clupea harengus* and *Micromesistius poutassou*. The former is a demersal spawner (Stratoudakis et al., 1998) and the latter performs diel vertical migrations and spends a large portion of the day-time near the bottom (Johnsen & Godø, 2007). Both these species

are also frequently caught by fishing gears targeting demersal fish assemblages (Afzali et al., 2021; Sousa et al., 2005).

2.2 | Data sources

Occurrence records for each species were obtained from two species occurrence data repositories, GBIF (<https://www.gbif.org/>) and OBIS (<https://www.obis.org/>; for all downloaded dataset citations, refer to Supporting Information Appendix S1). These were pruned to ensure that those included in the modelling framework matched the known distribution of each species and the time span of predictors (2000 and 2017; please refer to Supporting Information Appendix S2). Predictors were obtained from Bio-ORACLE, a dataset providing layers at 5 arcmin resolution (data processing detailed by Assis, Tyberghein, et al., 2018; Tyberghein et al., 2012); these included long-term maximum and minimum temperatures (MaxTemp and MinTemp; in degrees Celsius), long-term minimum salinity (Salinity; in practical salinity units), mean dissolved oxygen concentration (Oxygen; in moles per cubic metre) and mean primary productivity (Productivity; in grams per cubic metre per day). Predictors focused on the benthic realm (i.e., along the bottom) for near present-day conditions (long-term climatologies between 2000 and 2017) and the two contrasting scenarios of global change, RCP2.6 and RCP8.5. From the three benthic layers available per predictor in Bio-ORACLE, we chose to model species distributions with the

TABLE 1 Modelled species, their known depth range (in metres) and performance of boosted regression trees (BRT) and adaptive boosting (AdaBoost) distribution models in cross-validation and when ensembled in a final predictive model

Scientific name	Common name	Depth range (m)	Cross-validation	Final ensemble	
			AUC (BRT/AdaBoost)	AUC	Sensitivity
<i>Ammodytes marinus</i>	Lesser sand eel	10–150 ^a	.92/.89	.97	.95
<i>Clupea harengus</i>	Atlantic herring	0–364 ^b	.82/.73	.87	.79
<i>Dicentrarchus labrax</i>	European seabass	0–100 ^c	.83/.80	.92	.95
<i>Gadus morhua</i>	Atlantic cod	0–600 ^d	.82/.73	.89	.89
<i>Melanogrammus aeglefinus</i>	Haddock	0–450 ^e	.86/.81	.93	.92
<i>Merluccius merluccius</i>	European hake	30–1075 ^f	.87/.85	.93	.92
<i>Micromesistius poutassou</i>	Blue whiting	150–3000 ^d	.84/.82	.92	.79
<i>Pleuronectes platessa</i>	European plaice	0–200 ^e	.89/.86	.95	.97
<i>Solea solea</i>	Common sole	0–150 ^e	.86/.82	.93	.88
<i>Raja clavata</i>	Thornback ray	0–700 ^g	.84/.81	.93	.91

Note: For more comprehensive information of model performance, please refer to the Supporting Information (Appendix S3). AUC = area under the curve.

^aWright et al. (2000).

^bCoad and Reist (2004).

^cFrimodt (1995).

^dCohen et al. (1990).

^eMuus and Nielsen (1999).

^fLloris et al. (2005).

^gRelini et al. (1999).

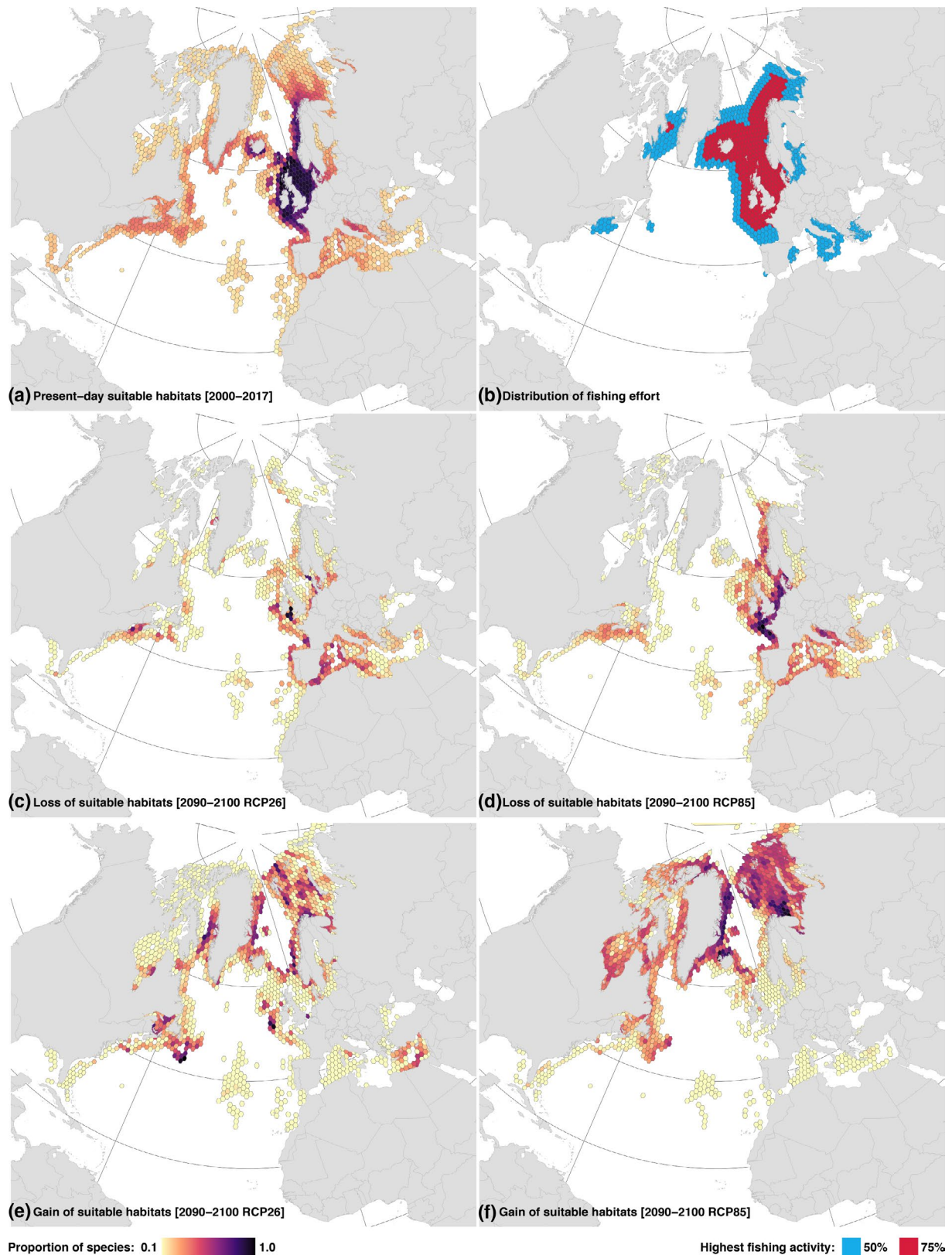


FIGURE 1 Proportion of near present-day suitable habitats, fishing effort and distributional range shifts [loss and gain of suitable habitats from near present-day conditions (2000–2017) to future Representative Concentration Pathway (RCP) climate change scenarios (2090–2100); for individual projections per species, please refer to Supporting Information Appendix S2]. Colour gradients reflect the proportion of species: (a) predicted with suitable habitats, and (c–f) that lost or gained suitable habitats (yellow to dark red depicting > 0 to 1 proportion scale). (b) The 50% (blue) and 75% (red) kernel utilization distribution of the fishing activity, by trawlers and demersal purse seiners, in the study area is represented

average depth layer. The RCP layers are provided as the ensemble of distinct models from the Coupled Model Intercomparison Project, namely, CCSM4, GFDL-ESM2G, HadGEM2-ES, IPSL-CM5A-LR and MIROC-ESM for temperatures and salinity, GFDL-ESM2G, HadGEM2-ES and IPSL-CM5A-LR for dissolved oxygen concentration and GFDL-ESM2G, IPSL-CM5A-LR and MIROC-ESM for primary productivity. Before modelling, Pearson's correlation coefficient was determined for all pairs of predictors, in addition to the variance inflation factor (VIF).

Gridded data of fishing effort from 2012 to 2016 (average hours per cell) was obtained from the Global Fishing Watch (Kroodsma et al., 2018) and filtered to include only fishing gear relevant to our focal species (demersal purse seiners and trawlers). These data were resampled using the nearest neighbour method (Hofstra et al., 2008) to match the resolution and spatial extent of predictors.

2.3 | Species distribution modelling

The distribution of species was modelled by ensembling (Araújo & New, 2007) the response of boosted regression trees (BRT) and adaptive boosting (AdaBoost), two machine learning algorithms with high predictive performances in SDMs (Assis et al., 2018; Krause-Jensen et al., 2020) that automatically fit complex interactions between predictor variables and nonlinear relationships, while reducing overfitting by optimal parametrization and monotonicity responses (Elith et al., 2008; Hofner et al., 2011). The algorithms fitted occurrence records per species (presences and pseudo-absences) against the predictor variables. Pseudo-absences were generated randomly in a 1:1 ratio with each set of species presence records, in locations where no presences were recorded (Barbet-Massin et al., 2012) and limited geographically to the biogeographical provinces (Spalding et al., 2007) where the species occur and their neighbouring provinces. This kept pseudo-absences in areas where dispersal might potentially occur, a crucial step to produce accurate SDMs (Barve et al., 2011). To eliminate excessive information, the correlation between predictors within the range of occurrence records was assessed as a function of geographical distance (Boavida et al., 2016). This was accomplished by building a correlogram to identify the shortest distance between occurrence records at which predictors were not correlated spatially. Records were then thinned by randomly selecting one record from the groups falling within correlated distances.

A cross-validation (CV) framework with 10-fold latitudinal bands was implemented to find the optimal hyperparameter combination to reduce overfitting and improve the potential for transferability of the model (Assis, Araújo, et al., 2018; Krause-Jensen et al., 2020; Vignali et al., 2020). The "grid search" method was used to test all possible hyperparameter combinations of the number of trees (50–1,000, step 50), tree complexity (1–6) and learning rate (.01, .005 and .001) for BRT, and the number of interactions (50–250, step 50), shrinkage (.25–1, step .25) and degrees of freedom (1–12) for AdaBoost (Fragkopoulou et al., 2021; Krause-Jensen et al., 2020). Models trained the different hyperparameters interactively by

fitting occurrence records with one withheld latitudinal band at a time, where performance was tested with the area under the curve (AUC) of the receiver operating characteristic curve (Fragkopoulou et al., 2021; Vignali et al., 2020). The optimal combination of hyperparameters was determined as the one producing models with the highest AUC in CV (Fragkopoulou et al., 2021; Krause-Jensen et al., 2020; Vignali et al., 2020). Overfitting was controlled further by forcing predictors to fit monotonic responses based on the biological effect (positive or negative) expected on the probability of occurrence of species (Hofner et al., 2011). MaxTemp fitted negative monotonic responses, whereas MinTemp, Salinity, Oxygen and Productivity fitted positive monotonic responses. The depth range, predictor choice and monotonic responses of species were set based on the literature (Supporting Information Appendix S1). The cross-validation framework also allowed estimation of the performance and potential transferability of the models in independent data, by reporting the average AUC obtained in CV when modelling with the optimal hyperparameters (Assis, Araújo, et al., 2018; Boavida et al., 2016; Fragkopoulou et al., 2021; Krause-Jensen et al., 2020; Wenger & Olden, 2012).

Final predictive BRT and AdaBoost models were trained with the full occurrence dataset, the predictor variables and the optimal hyperparameters found in cross-validation (Assis, Araújo, et al., 2018; Vignali et al., 2020). Predictive performance was also assessed with AUC and sensitivity (both metrics vary between zero and one) with models having AUC > .8 considered to have excellent performance and sensitivity describing the proportion of accurately predicted occurrences (Allouche et al., 2006; Hosmer & Lemeshow, 2000). The relative contribution of predictors to the models was assessed by computing the increase in AUC when each predictor was added to its alternative full model (i.e., the model including all predictors except that being tested). Wilcoxon rank tests inferred the significant increase of each predictor contribution when compared with its alternative full model.

Distribution maps representing habitat suitability per species were produced for near present-day conditions and the two global change scenarios with BRT and AdaBoost. These maps were reclassified to binomial presence-absence surfaces using a threshold maximizing both sensitivity and specificity (i.e., true negative rate; Assis, Araújo, et al., 2018; Jiménez-Valverde & Lobo, 2007). Climatic physiological limitations to species distributions (i.e., tipping points from which the models predict absence) were inferred by producing individual response functions for each predictor, while accounting for the average effect produced by all alternative predictors (Assis et al., 2017; Elith et al., 2008). Along each individual response function, tipping points were captured by extracting the extreme climatic values (maximum or minimum depending on the predictor) setting the transition between suitable and unsuitable habitat conditions, while considering the reclassified binomial predictive surface (Boavida et al., 2016).

2.4 | Latitudinal and depth distributional shifts

Latitudinal and depth distributional shifts were inferred by ensembling (mean function; Araújo & New, 2007) the maps produced with

BRT and AdaBoost for near present-day conditions and each of the RCP scenarios. The ensembles followed reclassification approach to binomial presence-absence surfaces using the maximization of sensitivity and specificity as detailed in the previous subsection. Distributional shifts were characterized by mirror density plots reflecting the potential latitudinal changes of species and by producing aggregated maps reflecting habitats gained and lost from near present-day conditions to future times.

The average and maximum predicted depth was determined per species by crossing a bathymetry layer obtained from Bio-ORACLE with the reclassified binomial maps. Depth shifts were assessed by comparing the median depth of species distributions in the present and future scenarios. Finally, the 50% and 75% kernel density of the distribution of fishing effort was estimated to assess whether areas of current high fishing effort might suffer future changes of suitable habitats for demersal fish. The assessment was made by comparing the proportion of species distributions in areas of high fishing activity in near-present day conditions and in future conditions.

3 | RESULTS

3.1 | Species distribution models

The distribution models using the optimal combination of hyperparameters (Supporting Information Appendix S3) produced predictions with high performance rates, both when tested in

cross-validation (average AUC in CV of BRT: $.86 \pm .03$; average AUC in CV of AdaBoost: $.82 \pm .04$; Table 1; Supporting Information Appendix S3) and when ensembled in unique models (average AUC: $.92 \pm .02$; Table 1). The ensembles reflected known distributional and depth ranges of each species (for maps contrasting occurrence records and predicted distributions, see Supporting Information Appendix S2). Focal fish species distributions were mostly predicted in the Celtic, Irish and North Seas and along the coasts of Norway and Iceland (Figure 1a).

The predictors considered in the models significantly increased predictive performances ($> .8$ proportion of improved models; Figure 2). Temperatures (i.e., maximum and minimum) alongside productivity were the main drivers explaining distributions, with a combined average contribution of 79.9% (TempMax: 27.9%; TempMin: 36.4%; Productivity: 15.6%; Figure 2a). Oxygen and Salinity made lower contributions (oxygen: 12.1%; salinity: 7.9%; Figure 2a); nevertheless, for some species, these predictors reached 23.0% and 17.2%, respectively (e.g., *Dicentrarchus labrax*). Additional contribution extremes/outliers were verified for *Micromesistius poutassou* (minimum temperature: 58.6%; salinity: -0.8% ; productivity: 1.4%) and *Ammodytes marinus* (productivity: 30.5%; for species-specific variable contributions, see Supporting Information Appendix S3). These findings are supported by the overall low correlation found between predictors; only minimum and maximum temperatures showed a stronger correlation (Pearson's correlation $.82$; VIF > 5 ; Supporting Information Appendix S4, Figure S4a; Table S4), yet their opposite monotonic fit in both BRT and AdaBoost (positive for

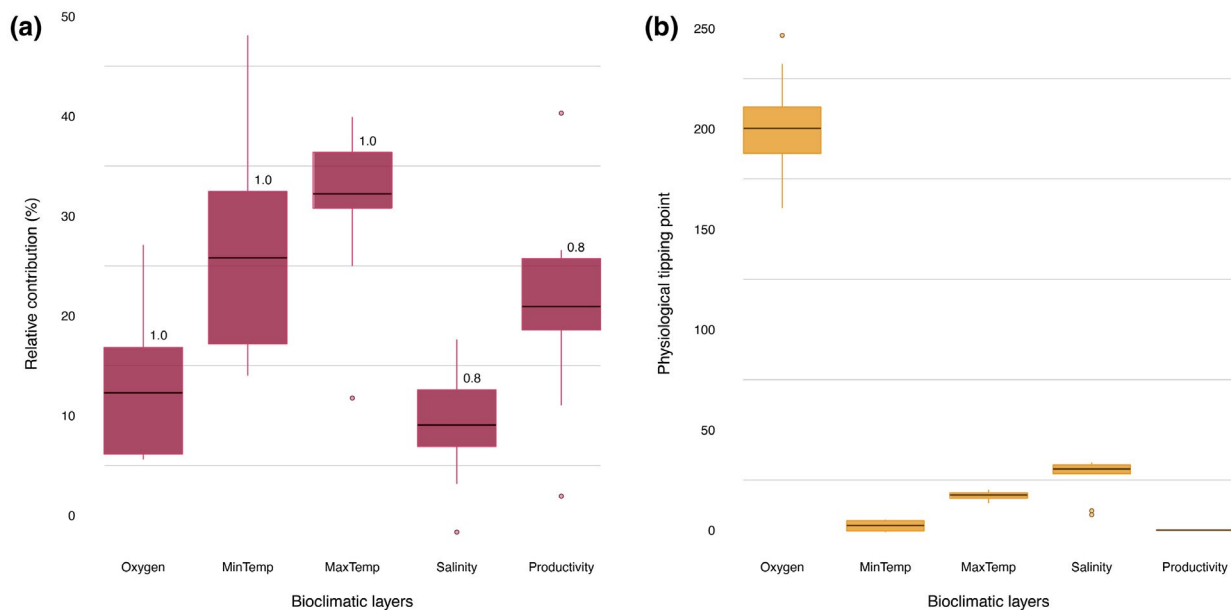


FIGURE 2 Average (a) relative contribution and (b) physiological tipping point of each predictor inferred with boosted regression trees and adaptive boosting across all species. Horizontal lines represent the mean contribution and tipping points (i.e., physiological tolerances) of each predictor. Values above boxes indicate the proportion of models developed for each species where predictors retrieved significantly higher contributions to their performance. Dissolved oxygen (Oxygen) is given in moles per cubic metre; minimum and maximum temperature (MinTemp and MaxTemp) in degrees Celsius; salinity (Salinity) in practical salinity units; and primary productivity (Productivity) in grams per cubic metre per day

minimum temperature and negative for maximum temperature) allowed removal of potential confounding inferences about the contributions of both predictors.

Focal species were predicted to find suitable habitat conditions in areas with near present-day average minimum temperatures between 0.1 and 5.4°C, maximum temperatures ranging from 13.6 to 20.2°C, dissolved oxygen concentrations inside the range 160.4–248.3 mol/m and salinity between 7.8 and 33.9 PSS. Productivity tipping points were found above oligotrophic conditions of 0–0.1 g/m³/day (Figure 2b; for species-specific variable tipping points, see Supporting Information Appendix S3).

3.2 | Latitudinal and depth distributional shifts

All species were projected to shift distributions poleward in both future scenarios, with greater changes in suitable areas (Figure 1c–f; Table 2) and latitudinal limits (Figure 3) when considering RCP8.5 (average range gain: $54.5 \pm 35.7\%$; average range loss: $40.9 \pm 13.4\%$) than with RCP2.6 (average range gain: $17.9 \pm 11.8\%$; average range loss: $13.4 \pm 4.1\%$). Latitudinal shifts were often coupled by depth shifts to deeper waters, which were also greater with RCP8.5 (average depth gain = 25.4 ± 21.5 m; 7 out of 10 species shift to deeper waters) than with RCP2.6 (average depth gain = 10.4 ± 7.9 m; 8 out of 10 of species shift to deeper waters; Figure 3).

In terms of suitable habitats, *C. harengus*, *Gadus morhua*, *Melanogrammus aeglefinus* and *Micromesistius poutassou* had the greatest range gain, while being among those with lower range loss (Table 2). These species have the potential to colonize coastal and open waters in the Barents, Kara, Norwegian, Greenland and Iceland Seas, the Hudson Bay and Western Arctic Ocean (for species-specific distribution projections under all scenarios analysed, see Supporting Information Appendix S2). On the contrary, *A. marinus*, *D. labrax*, *Merluccius merluccius*, *Pleuronectes platessa*, *Raja clavata* and *Solea solea* had smaller range gain and greater loss (with loss being concentrated in the southern section of their ranges). These species mostly colonize coastal waters in Norwegian, Iceland and Greenland Seas (Supporting Information Appendix S2). Changes in latitudinal range were a result of the colonization of the northernmost regions of the study area (especially for species with greater range gain) and the loss of southern suitable range (particularly evident for species with greater range loss; Figure 3).

Loss of suitable range is concentrated mostly in the Mediterranean, Celtic and Irish Seas, southern areas of the North Sea (especially along the English Channel) and on the north-east North American coast (Figure 1c,d). Poleward shifts are projected mostly towards the Barents, Greenland, Kara, Norwegian and Iceland Seas, Hudson Bay and the Western Arctic Ocean (Figure 1e,f). The geographical patterns of where species might lose or gain suitable areas are similar in both scenarios, but more pronounced for RCP8.5 (Figure 1c–f; Table 2).

Areas of high fishing effort are predominantly distributed in the north-east Atlantic, particularly in the Celtic, North, Iceland,

Norwegian and Barents Seas (Figure 1b). This matches regions where species are predicted to be distributed in near present-day conditions (suitable habitats overlapping areas of high fishing effort: $FE50 = 61.75 \pm 16.54\%$; Table 2) and projected in the future, (suitable habitats overlap ranging from 52.16 ± 27.16 to $79.36 \pm 16.19\%$, depending on the climate scenario and fishing effort; Table 2). The projected distributional changes in areas of high fishing effort are more pronounced in RCP85 than in RCP26 (Table 2), and the gains in suitable habitats in these areas (i.e., in Iceland, Norwegian and Barents Seas) surpass the projected losses, regardless of the climate scenario (i.e., in Celtic and North Seas; Figure 1c–f; Table 2).

4 | DISCUSSION

In the present study, we anticipate future distributional shifts for demersal fish of high commercial value in the northern Atlantic Ocean. On rare occasions, past studies have modelled species distributions based on predictors beyond temperature (e.g., Jones & Cheung, 2015); however, to our knowledge, this is the first attempt to integrate biologically meaningful predictors, such as oxygen, primary productivity and salinity, alongside temperature conditions, to assess the potential consequences of future climate changes to marine fish biodiversity at such large scales. Along the lines of previous research looking at the response of species to past and future climate change (Morato et al., 2020; Pinsky et al., 2020), our models predicted generalized poleward shifts across all modelled species. As hypothesized, greater changes to species distributions (i.e., suitable areas, latitudinal and depth distributions) were predicted under the higher greenhouse gas emission scenario (RCP8.5), suggesting that broad compliance with the Paris Agreement might potentially mitigate the negative impacts of global climate change to fish biodiversity, benefitting commercial fisheries and resource management (Baudron et al., 2020; Bell et al., 2015; Miller & Munro, 2004).

Overall, the most important predictors shaping demersal fish distributions (i.e., extreme temperatures and productivity) are in line with previous studies (Valavanis et al., 2004). Temperature directly impacts the physiology of species, and the modelled species, with contrasting distributional patterns, exhibited specific thermal limits and predictor affinities. Minimum temperature, for instance, contributed more than average to *Micromesistius poutassou*, a species with its northern range bordering the Arctic waters, where extremely cold conditions potentially impose an important physiological barrier. On the contrary, the lower contribution of maximum temperature to *D. labrax* results from its affinity for warmth (Beare et al., 2004), allowing the species to extend its range to the low-latitude limits of our study area. Besides temperature-dependent niche affinities, fish species favour high-productivity waters (Cheung et al., 2008), and this was evident for most modelled species (productivity tolerance limits above oligotrophic conditions). In particular, *A. marinus* relied greatly on this variable, because its distribution comprises exclusively the productive waters of the North and Irish Seas and the inner seas of the west coast of Scotland (Capuzzo et al., 2018). Beyond

TABLE 2 Near present-day (from 2000 to 2017) total area (in square kilometres) and percentage of species distributions inside areas of high fishing effort (FE50 and FE75); and RCP2.6 and RCP8.5 (from 2090 to 2100) percentage of lost and gained habitats in each future global change scenario and percentage of future distribution inside areas of high fishing effort

Species	Near present-day				RCP2.6				RCP8.5			
	Area ($\times 10^3$ km ²)	Percentage FE50	Percentage FE75	Area loss (%)	Area gain (%)	Percentage FE50	Percentage FE75	Area loss (%)	Area gain (%)	Percentage FE50	Percentage FE75	Percentage FE75
<i>Ammodytes marinus</i>	1,014.39	90.75	99.06	13.44	6.86	88.67 (−2.08)	99.35 (+0.29)	74.7	20.61	60.57 (−30.18)	93.02 (−6.04)	
<i>Clupea harengus</i>	4,393.32	48.14	74.21	13.01	31.78	39.43 (−8.71)	64.98 (−9.23)	36.64	113.46	17.57 (−30.57)	37.22 (−36.99)	
<i>Dicentrarchus labrax</i>	1,170.77	67.92	85.92	9.67	8.1	75.91 (+7.99)	90.55 (+4.63)	30.81	27.64	80.91 (+12.99)	94.85 (+8.93)	
<i>Gadus morhua</i>	4,993.17	36.97	61.28	6.64	32.51	27.79 (−9.18)	49.82 (−11.45)	29.77	75.25	14.39 (−22.58)	33.74 (−27.54)	
<i>Melanogrammus aeglefinus</i>	3,394.86	49.91	72.65	9.39	25.2	41.25 (−8.66)	65.82 (−6.83)	30.57	103.21	21.44 (−28.47)	40.52 (−32.13)	
<i>Merluccius merluccius</i>	2,051.04	57.72	78.12	20.14	10.88	68.49 (+10.77)	86.02 (+7.9)	37.73	30.37	78.33 (+20.61)	92.71 (+14.59)	
<i>Micromesistius poutassou</i>	4,392.44	44.82	66.54	18.92	37.7	40.55 (−4.27)	62.09 (−4.44)	33.24	90.21	33.25 (−11.57)	55.36 (−11.18)	
<i>Pleuronectes platessa</i>	1,624.06	80.31	94.76	13.45	9.76	78.52 (−1.79)	90.18 (−4.57)	55.37	41.8	53.22 (−27.09)	81.26 (−13.5)	
<i>Raja clavata</i>	2,066.72	62.78	79.44	16.89	10.91	73.8 (11.02)	88.35 (+8.92)	42.02	33.45	70.78 (+8)	87.03 (+7.59)	
<i>Solea solea</i>	1,323.13	78.11	93.43	12.25	5.13	83.67 (5.56)	96.48 (+3.05)	37.71	8.49	91.19 (+13.08)	98.88 (+5.45)	
Average	–	61.75	80.54	13.38	17.88	61.81	79.36	40.86	54.45	52.16	71.46	
SD	–	16.54	11.9	4.06	11.82	20.97	16.19	13.35	35.67	27.16	25.23	

Note: FE50 and FE75 represent 50% and 75% kernel utilization distribution of fishing effort, respectively. Bold values depict area changes > 50%, and values in parentheses show the difference in the percentage of species distributions inside areas of high fishing effort between near present-day and each future scenario.

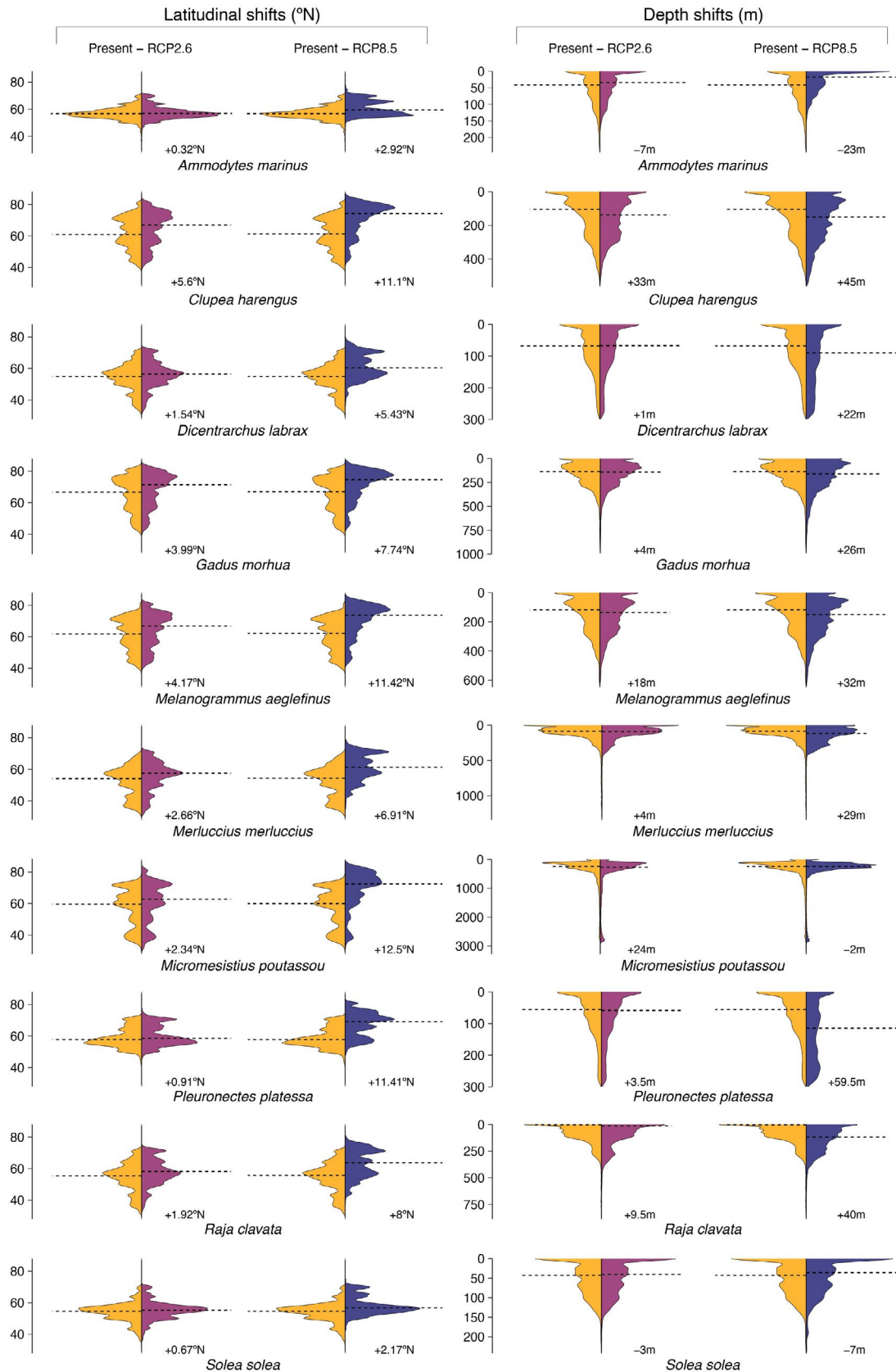


FIGURE 3 Latitudinal and depth shifts projected for demersal fish in the northern Atlantic with the Representative Concentration Pathway (RCP) scenarios RCP2.6 and RCP8.5 (future scenarios cover the time period 2090–2100). Yellow represents near present-day distributions; purple and blue represent distributions under RCP2.6 and RCP8.5, respectively. Dashed lines represent the median latitude and median depth of occurrence

these areas, productivity decreases sharply, as does the known and predicted distribution of this species. In contrast, productivity made a lower contribution for *Micromesistius poutassou*. In tandem with its wide-ranging distribution in the northern Atlantic Ocean, this species is also found in the Mediterranean Sea, a region known for poor nutrient/productivity conditions (Huertas et al., 2012). Salinity and dissolved oxygen were also identified as important drivers to model demersal fish species. Salinity controls osmoregulatory mechanisms, with areas falling outside tolerance limits presenting sharp distributional barriers (Evans & Claiborne, 2009). This premise was verified for the species with salinity tolerance limits > 28 PSS and absent from the Baltic Sea, the widest hyposaline region of the study area (Bagdonas et al., 2011). The more tolerant species, *P. platessa* and *C. harengus* (tolerance limits of 7.66 and 9.66 PSS, respectively), can expand their ranges into such conditions. Low-oxygen barriers prevail where the oxygen demand of species and their capacity for extraction and metabolic suppression exceed local oxygen availability (Pörtner & Knust, 2007). Overall, the inferred tolerance limits largely match known thresholds of hypoxia for marine fish (average of c. 200 mol/m³; Vaquer-Sunyer & Duarte, 2008), highlighting the key role of this variable for marine fish and strengthening the ecological significance of our models.

Distribution models reflecting meaningful biological traits showed the species favouring cold and eutrophic conditions that might change in the years to come. In particular, the critically identified regions with potential habitat losses (i.e., the Mediterranean Sea, the Celtic and Irish Seas, the southern portion of the North Sea and the north-east coast of North America) have previously been signposted for faster warming rates and susceptibility to future losses across biodiversity (Dulvy et al., 2008; Lejeune et al., 2010; Saba et al., 2016; Weinert et al., 2016). Projected latitudinal shifts were also coupled with depth shifts, as hypothesized. Similar to elevational shifts in terrestrial climates, stable deeper regions, less prone to abrupt warming when compared with superficial waters, seem to be crucial refugia for species to cope successfully with future climate conditions (Jorda et al., 2020). However, for some of the modelled species, distributional shifts produced shallower ranges, stemming from the loss of suitable deeper areas in near present-day conditions. This is likely to occur in deep areas of low-latitude ranges where near present-day environmental conditions already border physiological tolerances (Osovitz & Hofmann, 2007).

Ensemble model projections showed how higher-emission scenarios might translate into synergistic climate interactions linked to global warming, and greater distributional changes for demersal fish species. For instance, a reduction in dissolved oxygen is associated with warming conditions (Pörtner & Knust, 2007). This is particularly evident in projections along the coastlines of the northern Atlantic Ocean (Supporting Information Appendix S4, Figure S4d), acting synergistically with temperature in creating unsuitable habitats. Salinity is also projected to change in the years to come, with some regions impacted by an overall increase in salinity (Saba et al., 2016), whereas others are expected to experience reductions in salinity owing to freshwater runoff from melting ice (Belkin et al., 1998). The

spatial variation of this driver in the future ocean is likely to pose great challenges to species distributional ranges. In terms of primary productivity, projections show an increase in coastal waters (Supporting Information Appendix S4, Figure S4f), benefitting fish populations inhabiting regions where future temperatures do not surpass species-specific tolerance limits.

The biodiversity impact assessment we have produced has an inherent degree of uncertainty and oversimplification that should be acknowledged and discussed. First, it did not account for biotic interactions that might impact the capacity of species to shift distributions and colonize new suitable areas successfully (Selden et al., 2018). Neglecting this component might have led to overestimation of the potential of species to colonize new habitats in previously unsuitable areas. Second, it did not consider the ability of species to adapt to new conditions imposed by climate change. Although adaptation to new conditions has much relevance in the scope of anticipating climate-induced distributional shifts, it is highly unlikely to take place within the time frame considered (the next 80 years; Henson et al., 2017; Pinsky et al., 2020). Third, other potentially meaningful predictors, such as the sea bottom substrate (e.g., Koubbi et al., 2006; Maravelias et al., 2000; Maxwell et al., 2009), were not included in the SDM framework owing to their unavailability at the spatial scale of this study. This might have led to overestimation of predicted distributions; hence, our results should be interpreted as estimates of abiotic niche conditions rather than fully realized niches. Finally, given the large spatial scale of the present study, it is also possible that local conditions were not captured completely by AOGCMs, such as upwelling areas structuring future climate refugia (Lourenço et al., 2016). Nevertheless, the models mimicked species distributions and physiological tolerances, while integrating the best bioclimatic data available beyond thermal conditions alone.

Minimal climate change mitigation translated into greater environmental changes (RCP8.5) might produce larger negative consequences for demersal fish and the fisheries industry as a whole. Such is the case because more severe poleward shifts associated with greater environmental changes will be reflected in an increased need for investment in new fishing gear to adapt to new fishing conditions (Poloczanska, 2013; Sumaila et al., 2011) and will increase fishing trip costs to reach fishing grounds further north when compared with the potential cost associated with contained poleward shifts (Weatherdon et al., 2016). Coincidentally, areas that currently have higher fishing effort were among those predicted to suffer greater species turnover in the future (area losses: Celtic and North Seas; area gain: Iceland, Norwegian and Barents Seas). By limiting distributional shifts, the need for fishing adaptation and changes in distribution of effort is minimized. Furthermore, it allows species to allocate resources more efficiently to tackle additional threats alongside current climate change (e.g., overfishing and invasive processes). For instance, the combined effects of climate change, high fishing pressure and invasive processes have already caused great biodiversity loss in the Mediterranean Sea (Marbà et al., 2015; Occhipinti-Ambrogi, 2007). Together, our results emphasize the potential

benefits of complying with the Paris Agreement, limiting global warming levels (Wang et al., 2018), and therefore, range shifts in commercially valuable demersal fish. Nonetheless, in either climate change scenario, adaptive measures, such as redistribution of fishing effort and fishing stock allocation, are needed to reach sustainable exploitation of fish stocks and achieve the Sustainable Development Goals 14 targets (United Nations General Assembly, 2015). When rearranging the distribution of fishing effort, managers should strive to guide fishers to areas where targeted species are found, allowing for greater yield per fishing trip. Additionally, the allocation of fishing stocks per country should reflect species distributions at the time of allocation and their vulnerability to environmental conditions. Lack of consideration of climate change-induced range shifts has already caused overfishing, massive discarding of fished volume and conflicts between neighbouring regions owing to poor allocation of fishing effort and fish stocks (Baudron et al., 2020; Baudron & Fernandes, 2015). Conservation planning strategies should consider climate change impacts to achieve their maximum effectiveness. Marine spatial planning, and Marine Protected Areas in particular, are promising conservation tools; nevertheless, the inclusion of climate change threats is still rare, which reduces their potential effectiveness (Rilov et al., 2019). Accurate predictions that truly grasp the physiological limitations of species, such as the ones presented here, are key to provide a reliable, policy-relevant biodiversity impact assessment (e.g., IPCC) and contribute with baseline projections for the distribution of fishing stocks and allocation of fishing effort.

ACKNOWLEDGMENTS

This study was supported by the European Maritime and Fisheries Fund, the MAR2020 program through project REDAMP (MAR-01.04.02-FEAMP-0015), and by the Foundation for Science and Technology (FCT) of Portugal through the projects UID/Multi/04326/2019 and PTDC/BIA-CBI/6515/2020 and the transitional norm DL57/2016/CP1361/CT0035 and DL57/2016/CP1361/CT0036. All outlined data providers were also crucial to the execution of the present paper.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Figshare at: <https://doi.org/10.6084/m9.figshare.14519589>

ORCID

Manuel Ramos Martins  <https://orcid.org/0000-0001-6477-2518>

Jorge Assis  <https://orcid.org/0000-0002-6624-4820>

David Abecasis  <https://orcid.org/0000-0002-9802-8153>

REFERENCES

- Afzali, S. F., Bourdages, H., Laporte, M., Mérot, C., Normandeau, E., Audet, C., & Bernatchez, L. (2021). Comparing environmental metabarcoding and trawling survey of demersal fish communities in the Gulf of St. Lawrence, Canada. *Environmental DNA*, 3, 22–42. <https://doi.org/10.1002/edn3.111>
- Allouche, O., Tsoar, A., & Kadmon, R. (2006). Assessing the accuracy of species distribution models: Prevalence, kappa and the true skill statistic (TSS). *Journal of Applied Ecology*, 43, 1223–1232. <https://doi.org/10.1111/j.1365-2664.2006.01214.x>
- Araújo, M. B., & New, M. (2007). Ensemble forecasting of species distributions. *Trends in Ecology and Evolution*, 22, 42–47. <https://doi.org/10.1016/j.tree.2006.09.010>
- Assis, J., Araújo, M. B., & Serrão, E. A. (2018). Projected climate changes threaten ancient refugia of kelp forests in the North Atlantic. *Global Change Biology*, 24, e55–e66. <https://doi.org/10.1111/gcb.13818>
- Assis, J., Bercebar, E., Claro, B., Alberto, F., Reed, D., Raimondi, P., & Serrão, E. A. (2017). Major shifts at the range edge of marine forests: The combined effects of climate changes and limited dispersal. *Scientific Reports*, 7, 1–10. <https://doi.org/10.1038/srep44348>
- Assis, J., Tyberghein, L., Bosch, S., Verbruggen, H., Serrão, E. A., & De Clerck, O. (2018). Bio-ORACLE v2.0: Extending marine data layers for bioclimatic modelling. *Global Ecology and Biogeography*, 27, 277–284. <https://doi.org/10.1111/geb.12693>
- Bagdonas, K., Nika, N., Bristow, G., Jankauskiene, R., Salyte, A., & Kontautas, A. (2011). First record of *Dicentrarchus labrax* (Linnaeus, 1758) from the southeastern Baltic Sea (Lithuania). *Journal of Applied Ichthyology*, 27, 1390–1391. <https://doi.org/10.1111/j.1439-0426.2011.01817.x>
- Barbet-Massin, M., Jiguet, F., Albert, C. H., & Thuiller, W. (2012). Selecting pseudo-absences for species distribution models: How, where and how many? *Methods in Ecology and Evolution*, 3, 327–338. <https://doi.org/10.1111/j.2041-210X.2011.00172.x>
- Barve, N., Barve, V., Jiménez-Valverde, A., Lira-Noriega, A., Maher, S. P., Peterson, A. T., Soberón, J., & Villalobos, F. (2011). The crucial role of the accessible area in ecological niche modeling and species distribution modeling. *Ecological Modelling*, 222, 1810–1819. <https://doi.org/10.1016/j.ecolmodel.2011.02.011>
- Baudron, A. R., Brunel, T., Blanchet, M., Hidalgo, M., Chust, G., Brown, E. J., Kleisner, K. M., Millar, C., MacKenzie, B. R., Nikolioudakis, N., Fernandes, J. A., & Fernandes, P. G. (2020). Changing fish distributions challenge the effective management of European fisheries. *Ecography*, 43, 494–505. <https://doi.org/10.1111/ecog.04864>
- Baudron, A. R., & Fernandes, P. G. (2015). Adverse consequences of stock recovery: European hake, a new “choke” species under a discard ban? *Fish and Fisheries*, 16, 563–575. <https://doi.org/10.1111/faf.12079>
- Beare, D. J., Burns, F., Greig, A., Jones, E. G., Peach, K., Kienzie, M., McKenzie, E., & Reid, D. G. (2004). Long-term increases in prevalence of North Sea fishes having southern biogeographic affinities. *Marine Ecology Progress Series*, 284, 269–278. <https://doi.org/10.3354/meps284269>
- Belkin, I. M., Levitus, S., Antonov, J., & Malmberg, S. A. (1998). “Great Salinity Anomalies” in the North Atlantic. *Progress in Oceanography*, 41, 1–68. [https://doi.org/10.1016/S0079-6611\(98\)00015-9](https://doi.org/10.1016/S0079-6611(98)00015-9)
- Bell, R. J., Richardson, D. E., Hare, J. A., Lynch, P. D., & Fratantoni, P. S. (2015). Disentangling the effects of climate, abundance, and size on the distribution of marine fish: An example based on four stocks from the Northeast US shelf. *ICES Journal of Marine Science*, 72, 1311–1322. <https://doi.org/10.1093/icesjms/fsu217>
- Boavida, J., Assis, J., Silva, I., & Serrão, E. A. (2016). Overlooked habitat of a vulnerable gorgonian revealed in the Mediterranean and Eastern Atlantic by ecological niche modelling. *Scientific Reports*, 6, 1–9. <https://doi.org/10.1038/srep36460>
- Breitbart, D., Levin, L. A., Oschlies, A., Grégoire, M., Chavez, F. P., Conley, D. J., Garçon, V., Gilbert, D., Gutiérrez, D., Isensee, K., Jacinto, G. S., Limburg, K. E., Montes, I., Naqvi, S. W. A., Pitcher, G. C., Rabalais, N. N., Roman, M. R., Rose, K. A., Seibel, B. A., ... Zhang, J. (2018). Declining oxygen in the global ocean and coastal waters. *Science*, 359, eaam7240. <https://doi.org/10.1126/science.aam7240>

- Capuzzo, E., Lynam, C. P., Barry, J., Stephens, D., Forster, R. M., Greenwood, N., McQuatters-Gollop, A., Silva, T., van Leeuwen, S. M., & Engelhard, G. H. (2018). A decline in primary production in the North Sea over 25 years, associated with reductions in zooplankton abundance and fish stock recruitment. *Global Change Biology*, 24, e352–e364. <https://doi.org/10.1111/gcb.13916>
- Champion, C., Hobday, A. J., Tracey, S. R., & Pecl, G. T. (2018). Rapid shifts in distribution and high-latitude persistence of oceanographic habitat revealed using citizen science data from a climate change hotspot. *Global Change Biology*, 24, 5440–5453. <https://doi.org/10.1111/gcb.14398>
- Cheung, W. W. L., Close, C., Lam, V., Watson, R., & Pauly, D. (2008). Application of macroecological theory to predict effects of climate change on global fisheries potential. *Marine Ecology Progress Series*, 365, 187–197. <https://doi.org/10.3354/meps07414>
- Cheung, W. W. L., Lam, V. W. Y., Sarmiento, K. L., Kearney, K., Watson, R., Zeller, D., & Pauly, D. (2010). Large-scale redistribution of maximum fisheries catch potential in the global ocean under climate change. *Global Change Biology*, 16, 24–35. <https://doi.org/10.1111/j.1365-2486.2009.01995.x>
- Cheung, W. W. L., Watson, R., & Pauly, D. (2013). Signature of ocean warming in global fisheries catch. *Nature*, 497, 365–368. <https://doi.org/10.1038/nature12156>
- Coad, B. W., & Reist, J. D. (2004). Annotated list of the Arctic marine fishes of Canada. *Canadian Manuscript Report of Fisheries and Aquatic Sciences*, 2674, 112.
- Cohen, D., Inada, T., Iwamoto, T., & Scialabba, N. (1990). Gadiform fishes of the world (Order Gadiformes): an annotated and illustrated catalogue of cods, hakes, grenadiers and other Gadiform fishes known to date. In W. Fischer W. Schneider & L. Garibaldi (Eds.), *FAO fisheries synopsis* (Vol. 125, Issue 10). FAO.
- Diaz, R. J., & Rosenberg, R. (2008). Spreading dead zones and consequences for marine ecosystems. *Science*, 321, 926–929. <https://doi.org/10.1126/science.1156401>
- Dulvy, N. K., Rogers, S. I., Jennings, S., Stelzenmüller, V., Dye, S. R., & Skjoldal, H. R. (2008). Climate change and deepening of the North Sea fish assemblage: A biotic indicator of warming seas. *Journal of Applied Ecology*, 45, 1029–1039. <https://doi.org/10.1111/j.1365-2664.2008.01488.x>
- Duncan, M. I., James, N. C., Potts, W. M., & Bates, A. E. (2020). Different drivers, common mechanism; the distribution of a reef fish is restricted by local-scale oxygen and temperature constraints on aerobic metabolism. *Conservation Physiology*, 8, coaa090. <https://doi.org/10.1093/conphys/coaa090>
- Elith, J., Leathwick, J. R., & Hastie, T. (2008). A working guide to boosted regression trees. *Journal of Animal Ecology*, 77, 802–813. <https://doi.org/10.1111/j.1365-2656.2008.01390.x>
- EU. (2018). *Facts and figures on the common fisheries policy. Basic statistical data* (2018th ed.). European Commission.
- Evans, D. H., & Claiborne, J. B. (2009). Osmotic and ionic regulation in fishes. In D. H. Evans (Ed.), *Osmotic and ionic regulation: Cells and animals*. CRC Press.
- FAO Fisheries Department. (2011). *Review of the state of world marine fishery resources 2011. Marine resources—Northwest Atlantic*. Fisheries and Resources Monitoring System (FIRMS).
- Fragkopoulou, E., Serrão, E. A., Horta, P. A., Koerich, G., & Assis, J. (2021). Bottom trawling threatens future climate refugia of rhodoliths globally. *Frontiers in Marine Science*, 7, 1246.
- Free, C. M., Thorson, J. T., Pinsky, M. L., Oken, K. L., Wiedenmann, J., & Jensen, O. P. (2019). Impacts of historical warming on marine fisheries production. *Science*, 363, 979–983. <https://doi.org/10.1126/science.aau1758>
- Frimodt, C. (1995). *Multilingual illustrated guide to the world's commercial warmwater fish*. Fishing News Books.
- Geden, O. (2016). The Paris agreement and the inherent inconsistency of climate policy making. *Wires Climate Change*, 7, 790–797. <https://doi.org/10.1002/wcc.427>
- Handorf, D., & Dethloff, K. (2012). How well do state-of-the-art atmosphere-ocean general circulation models reproduce atmospheric teleconnection patterns? *Tellus A: Dynamic Meteorology and Oceanography*, 64, 19777. <https://doi.org/10.3402/tellusa.v64i0.19777>
- Henson, S. A., Beaulieu, C., Ilyina, T., John, J. G., Long, M., Séférián, R., Tjiputra, J., & Sarmiento, J. L. (2017). Rapid emergence of climate change in environmental drivers of marine ecosystems. *Nature Communications*, 8, 14682. <https://doi.org/10.1038/ncomm14682>
- Hofner, B., Müller, J., & Hothorn, T. (2011). Monotonicity-constrained species distribution models. *Ecology*, 92, 1895–1901. <https://doi.org/10.1890/10-2276.1>
- Hofstra, N., Haylock, M., New, M., Jones, P., & Frei, C. (2008). Comparison of six methods for the interpolation of daily, European climate data. *Journal of Geophysical Research Atmospheres*, 113, 1–19. <https://doi.org/10.1029/2008JD010100>
- Hosmer, D. W., & Lemeshow, S. (2000). Chapter 5 - Assessing the Fit of the Model N. Cressie N. Fisher I. Johnstone J. Kadane D. Scott B. Silverman A. Smith & J. Teugels (Eds.), *Applied logistic regression* (2nd ed., pp. 160–164). John Wiley and Sons.
- Huertas, I. E., Ríos, A. F., García-Lafuente, J., Navarro, G., Makaoui, A., Sanchez-Romn, A., Rodriguez-Galvez, S., Orbi, A., Ruiz, J., & Pérez, F. F. (2012). Atlantic forcing of the Mediterranean oligotrophy. *Global Biogeochemical Cycles*, 26, GB2022. 2022-. <https://doi.org/10.1029/2011GB004167>
- Jiménez-Valverde, A., & Lobo, J. M. (2007). Threshold criteria for conversion of probability of species presence to either-or presence-absence. *Acta Oecologica*, 31, 361–369. <https://doi.org/10.1016/j.actao.2007.02.001>
- Johnsen, E., & Godø, O. R. (2007). Diel variations in acoustic recordings of blue whiting (*Micromesistius poutassou*). *ICES Journal of Marine Science*, 64, 1202–1209. <https://doi.org/10.1093/icesjms/fsm110>
- Jones, M. C., & Cheung, W. W. L. (2015). Multi-model ensemble projections of climate change effects on global marine biodiversity. *ICES Journal of Marine Science*, 72, 741–752. <https://doi.org/10.1093/icesjms/fsu172>
- Jorda, G., Marbà, N., Bennett, S., Santana-Garcon, J., Agusti, S., & Duarte, C. M. (2020). Ocean warming compresses the three-dimensional habitat of marine life. *Nature Ecology and Evolution*, 4, 109–114. <https://doi.org/10.1038/s41559-019-1058-0>
- Kleisner, K. M., Fogarty, M. J., McGee, S., Hare, J. A., Moret, S., Perretti, C. T., & Saba, V. S. (2017). Marine species distribution shifts on the U.S. Northeast Continental Shelf under continued ocean warming. *Progress in Oceanography*, 153, 24–36. <https://doi.org/10.1016/j.pocean.2017.04.001>
- Koubbi, P., Loots, C., Cotonnec, G., Harlay, X., Griocche, A., Vaz, S., Martin, C., Walkey, M., & Carpentier, A. (2006). Spatial patterns and GIS habitat modelling of *Solea solea*, *Pleuronectes flesus* and *Limanda limanda* fish larvae in the eastern English Channel during the spring. *Scientia Marina*, 70S2, 147–157.
- Krause-Jensen, D., Archambault, P., Assis, J., Bartsch, I., Bischof, K., Filbee-Dexter, K., Dunton, K. H., Maximova, O., Ragnarsdóttir, S. B., Sejr, M. K., Simakova, U., Spiridonov, V., Wegeberg, S., Winding, M. H. S., & Duarte, C. M. (2020). Imprint of climate change on pan-Arctic marine vegetation. *Frontiers in Marine Science*, 7, 1129.
- Kroodsma, D. A., Mayorga, J., Hochberg, T., Miller, N. A., Boerder, K., Ferretti, F., Wilson, A., Bergman, B., White, T. D., Block, B. A., Woods, P., Sullivan, B., Costello, C., & Worm, B. (2018). Tracking the global footprint of fisheries. *Science*, 359, 904–908. <https://doi.org/10.1126/science.aao5646>

- Lejeusne, C., Chevaldonné, P., Pergent-Martini, C., Boudouresque, C. F., & Pérez, T. (2010). Climate change effects on a miniature ocean: The highly diverse, highly impacted Mediterranean Sea. *Trends in Ecology and Evolution*, 25, 250–260. <https://doi.org/10.1016/j.tree.2009.10.009>
- Lloris, D., Matallanas, J., & Oliver, P. (2005). *Hakes of the world (Family Merlucciidae). An annotated and illustrated catalogue of hake species known to date* (Issue 2). FAO Species Catalogue for Fishery Purposes.
- Lourenço, C. R., Zardi, G. I., McQuaid, C. D., Serrão, E. A., Pearson, G. A., Jacinto, R., & Nicastro, K. R. (2016). Upwelling areas as climate change refugia for the distribution and genetic diversity of a marine macroalga. *Journal of Biogeography*, 43, 1595–1607. <https://doi.org/10.1111/jbi.12744>
- Maravelias, C. D., Reid, D. G., & Swartzman, G. (2000). Modelling spatio-temporal effects of environment on Atlantic herring, *Clupea harengus*. *Environmental Biology of Fishes*, 58, 157–172. <https://doi.org/10.1023/A:1007693732571>
- Marbà, N., Jordà, G., Agustí, S., Girard, C., & Duarte, C. M. (2015). Footprints of climate change on Mediterranean Sea biota. *Frontiers in Marine Science*, 2, 56. <https://doi.org/10.3389/fmars.2015.00056>
- Maxwell, D. L., Stelzenmüller, V., Eastwood, P. D., & Rogers, S. I. (2009). Modelling the spatial distribution of plaice (*Pleuronectes platessa*), sole (*Solea solea*) and thornback ray (*Raja clavata*) in UK waters for marine management and planning. *Journal of Sea Research*, 61, 258–267. <https://doi.org/10.1016/j.seares.2008.11.008>
- McHenry, J., Welch, H., Lester, S. E., & Saba, V. (2019). Projecting marine species range shifts from only temperature can mask climate vulnerability. *Global Change Biology*, 25, 4208–4221. <https://doi.org/10.1111/gcb.14828>
- Melo-Merino, S. M., Reyes-Bonilla, H., & Lira-Noriega, A. (2020). Ecological niche models and species distribution models in marine environments: A literature review and spatial analysis of evidence. *Ecological Modelling*, 415, 108837. <https://doi.org/10.1016/j.ecolmodel.2019.108837>
- Miller, K. A., & Munro, G. R. (2004). Climate and cooperation: A new perspective on the management of shared fish stocks. *Marine Resource Economics*, 19, 367–393. <https://doi.org/10.1086/mre.19.3.42629440>
- Mills, K. E., Pershing, A. J., Brown, C. J., Chen, Y., Chiang, F. S., Holland, D. S., Lehuta, S., Nye, J. A., Sun, J. C., Thomas, A. C., & Wahle, R. A. (2013). Fisheries management in a changing climate: Lessons from the 2012 ocean heat wave in the Northwest Atlantic. *Oceanography*, 26, 191–195. <https://doi.org/10.5670/oceanog.2013.27>
- Morato, T., González-Irusta, J.-M., Dominguez-Carrió, C., Wei, C.-L., Davies, A., Sweetman, A. K., Taranto, G. H., Beazley, L., García-Alegre, A., Grehan, A., Laffargue, P., Murillo, F. J., Sacau, M., Vaz, S., Kenchington, E., Arnaud-Haond, S., Callery, O., Chimienti, G., Cordes, E., ... Carreiro-Silva, M. (2020). Climate-induced changes in the suitable habitat of cold-water corals and commercially important deep-sea fishes in the North Atlantic. *Global Change Biology*, 26, 2181–2202. <https://doi.org/10.1111/gcb.14996>
- Muus, B. J., & Nielsen, J. G. (1999). *Sea fish*. Scandinavian Fishing Year Book.
- Occhipinti-Ambrogi, A. (2007). Global change and marine communities: Alien species and climate change. *Marine Pollution Bulletin*, 55, 342–352. <https://doi.org/10.1016/j.marpolbul.2006.11.014>
- Osovitz, C. J., & Hofmann, G. E. (2007). Marine macrophysiology: Studying physiological variation across large spatial scales in marine systems. *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology*, 147, 821–827. <https://doi.org/10.1016/j.cbpa.2007.02.012>
- Pecl, G. T., Araújo, M. B., Bell, J. D., Blanchard, J., Bonebrake, T. C., Chen, I.-C., Clark, T. D., Colwell, R. K., Danielsen, F., Evengård, B., Falconi, L., Ferrier, S., Frusher, S., García, R. A., Griffiths, R. B., Hobday, A. J., Janion-Scheepers, C., Jarzyna, M. A., Jennings, S., ... Williams, S. E. (2017). Biodiversity redistribution under climate change: Impacts on ecosystems and human well-being. *Science*, 355, eaai9214. <https://doi.org/10.1126/science.aai9214>
- Perry, A. L., Low, P. J., Ellis, J. R., & Reynolds, J. D. (2005). Climate change and distribution shifts in marine fishes. *Science*, 308, 1912–1915. <https://doi.org/10.1126/science.1111322>
- Pinsky, M. L., Selden, R. L., & Kitchel, Z. J. (2020). Climate-driven shifts in marine species ranges: Scaling from organisms to communities. *Annual Review of Marine Science*, 12, 14–32. <https://doi.org/10.1146/annurev-marine-010419-010916>
- Poloczanska, E. S., Brown, C. J., Sydeman, W. J., Kiessling, W., Schoeman, D. S., Moore, P. J., Brander, K., Bruno, J. F., Buckley, L. B., Burrows, M. T., Duarte, C. M., Halpern, B. S., Holding, J., Kappel, C. V., O'Connor, M. I., Pandolfi, J. M., Parmesan, C., Schwing, F., Thompson, S. A., & Richardson, A. J. (2013). Global imprint of climate change on marine life. *Nature Climate Change*, 3, 919–925. <https://doi.org/10.1038/nclimate1958>
- Pörtner, H. O., & Farrell, A. P. (2008). Physiology and climate change. *Science*, 322, 690–692.
- Pörtner, H. O., & Knust, R. (2007). Climate change affects marine fishes through the oxygen limitation of thermal tolerance. *Science*, 315, 95–97. <https://doi.org/10.1126/science.1135471>
- Relini, G., Bertrand, J., & Zamboni, A. (1999). Synthesis of the knowledge of bottom fishery resources in Central Mediterranean (Italy and Corsica). *Biologia Marina Mediterranea*, 6, 259–268.
- Riahi, K., Rao, S., Krey, V., Cho, C., Chirkov, V., Fischer, G., Kindermann, G., Nakicenovic, N., & Rafaj, P. (2011). RCP8.5—A scenario of comparatively high greenhouse gas emissions. *Climatic Change*, 109, 33–57. <https://doi.org/10.1007/s10584-011-0149-y>
- Rilov, G., Mazaris, A. D., Stelzenmüller, V., Helmuth, B., Wahl, M., Guy-Haim, T., Mieszowska, N., Ledoux, J. B., & Katsanevakis, S. (2019). Adaptive marine conservation planning in the face of climate change: What can we learn from physiological, ecological and genetic studies? *Global Ecology and Conservation*, 17, e00566. <https://doi.org/10.1016/j.gecco.2019.e00566>
- Rödger, D., & Lötters, S. (2010). Explanative power of variables used in species distribution modelling: An issue of general model transferability or niche shift in the invasive Greenhouse frog (*Eleutherodactylus planirostris*). *Naturwissenschaften*, 97, 781–796. <https://doi.org/10.1007/s00114-010-0694-7>
- Rutterford, L. A., Simpson, S. D., Jennings, S., Johnson, M. P., Blanchard, J. L., Schön, P. J., Sims, D. W., Tinker, J., & Genner, M. J. (2015). Future fish distributions constrained by depth in warming seas. *Nature Climate Change*, 5, 569–573. <https://doi.org/10.1038/nclimate2607>
- Saba, V. S., Griffies, S. M., Anderson, W. G., Winton, M., Alexander, M. A., Delworth, T. L., Hare, J. A., Harrison, M. J., Rosati, A., Vecchi, G. A., & Zhang, R. (2016). Enhanced warming of the Northwest Atlantic Ocean under climate change. *Journal of Geophysical Research: Oceans*, 121, 118–132.
- Selden, R. L., Batt, R. D., Saba, V. S., & Pinsky, M. L. (2018). Diversity in thermal affinity among key piscivores buffers impacts of ocean warming on predator–prey interactions. *Global Change Biology*, 24, 117–131. <https://doi.org/10.1111/gcb.13838>
- Sousa, P., Azevedo, M., & Gomes, M. C. (2005). Demersal assemblages off Portugal: Mapping, seasonal, and temporal patterns. *Fisheries Research*, 75, 120–137. <https://doi.org/10.1016/j.fishres.2005.03.012>
- Spalding, M. D., Fox, H. E., Allen, G. R., Davidson, N., Ferdaña, Z. A., Finlayson, M., Halpern, B. S., Jorge, M. A., Lombana, A., Lourie, S. A., Martin, K. D., McManus, E., Molnar, J., Recchia, C. A., & Robertson, J. (2007). Marine ecoregions of the world: A bioregionalization of coastal and shelf areas. *BioScience*, 57, 573–583. <https://doi.org/10.1641/B570707>

- Spence, A. R., & Tingley, M. W. (2020). The challenge of novel abiotic conditions for species undergoing climate-induced range shifts. *Ecography*, 43, 1571–1590. <https://doi.org/10.1111/ecog.05170>
- Stratoudakis, Y., Gallego, A., & Morrison, J. A. (1998). Spatial distribution of developmental egg ages within a herring *Clupea harengus* spawning ground. *Marine Ecology Progress Series*, 174, 27–32. <https://doi.org/10.3354/meps174027>
- Sumaila, U. R., Cheung, W. W. L., Lam, V. W. Y., Pauly, D., & Herrick, S. (2011). Climate change impacts on the biophysics and economics of world fisheries. *Nature Climate Change*, 1, 449–456. <https://doi.org/10.1038/nclimate1301>
- Sunday, J. M., Bates, A. E., & Dulvy, N. K. (2012). Thermal tolerance and the global redistribution of animals. *Nature Climate Change*, 2, 686–690.
- Tyberghein, L., Verbruggen, H., Pauly, K., Troupin, C., Mineur, F., & De Clerck, O. (2012). Bio-ORACLE: A global environmental dataset for marine species distribution modelling. *Global Ecology and Biogeography*, 21, 272–281. <https://doi.org/10.1111/j.1466-8238.2011.00656.x>
- United Nations Framework Convention on Climate Change. (2015). *Paris Agreement*. United Nations.
- United Nations General Assembly. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development*, 21 October 2015, A/RES/70/1. United Nations. <https://www.refworld.org/docid/57b6e3e44.html>
- Valavanis, V. D., Kapantagakis, A., Katara, I., & Palialexis, A. (2004). Critical regions: A GIS-based model of marine productivity hotspots. *Aquatic Sciences*, 66, 139–148. <https://doi.org/10.1007/s00027-003-0669-2>
- van Vuuren, D. P., Stehfest, E., den Elzen, M. G. J., Kram, T., van Vliet, J., Deetman, S., Isaac, M., Goldewijk, K. K., Hof, A., Beltran, A. M., Oostenrijk, R., & van Ruijven, B. (2011). RCP2.6: Exploring the possibility to keep global mean temperature increase below 2°C. *Climatic Change*, 109, 95–116. <https://doi.org/10.1007/s10584-011-0152-3>
- Vaquier-Sunyer, R., & Duarte, C. M. (2008). Thresholds of hypoxia for marine biodiversity. *Proceedings of the National Academy of Sciences USA*, 105, 15452–15457. <https://doi.org/10.1073/pnas.0803833105>
- Vignali, S., Barras, A. G., Arlettaz, R., & Braunisch, V. (2020). SDMtune: An R package to tune and evaluate species distribution models. *Ecology and Evolution*, 10, 11488–11506.
- Wang, F., Tokarska, K. B., Zhang, J., Ge, Q., Hao, Z., Zhang, X., & Wu, M. (2018). Climate warming in response to emission reductions consistent with the Paris agreement. *Advances in Meteorology*, 2018, 1–9. <https://doi.org/10.1155/2018/2487962>
- Warren, R., Price, J., VanDerWal, J., Cornelius, S., & Sohl, H. (2018). The implications of the United Nations Paris Agreement on climate change for globally significant biodiversity areas. *Climatic Change*, 147, 395–409. <https://doi.org/10.1007/s10584-018-2158-6>
- Weatherdon, L., Magnan, A. K., Rogers, A. D., Sumaila, U. R., & Cheung, W. W. L. (2016). Observed and projected impacts of climate change on marine fisheries, aquaculture, coastal tourism, and human health: An update. *Frontiers in Marine Science*, 3, 48. <https://doi.org/10.3389/fmars.2016.00048>
- Weinert, M., Mathis, M., Kröncke, I., Neumann, H., Pohlmann, T., & Reiss, H. (2016). Modelling climate change effects on benthos: Distributional shifts in the North Sea from 2001 to 2099. *Estuarine, Coastal and Shelf Science*, 175, 157–168. <https://doi.org/10.1016/j.ecss.2016.03.024>
- Wenger, S. J., & Olden, J. D. (2012). Assessing transferability of ecological models: An underappreciated aspect of statistical validation. *Methods in Ecology and Evolution*, 3, 260–267. <https://doi.org/10.1111/j.2041-210X.2011.00170.x>
- Wright, P. J., Jensen, H., & Tuck, I. (2000). The influence of sediment type on the distribution of the lesser sandeel, *Ammodytes marinus*. *Journal of Sea Research*, 44, 243–256. [https://doi.org/10.1016/S1385-1101\(00\)00050-2](https://doi.org/10.1016/S1385-1101(00)00050-2)

BIOSKETCH

Manuel Ramos Martins (research assistant), **Jorge Assis** (post-doctoral researcher) and **David Abecasis** (researcher) are based at CCMAR, University of Algarve. Their research is focused on ecological niche modelling, past and future climate-driven range shifts, landscape genetics at multiple temporal and spatial scales, planning/management of marine protected areas and marine fish ecology.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the Supporting Information section.

How to cite this article: Ramos Martins M, Assis J, Abecasis D. Biologically meaningful distribution models highlight the benefits of the Paris Agreement for demersal fishing targets in the North Atlantic Ocean. *Global Ecol Biogeogr*. 2021;00: 1–14. <https://doi.org/10.1111/geb.13327>